

Structural Estimation Under Misspecification

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Motivation and contribution

A *tightly parameterized structural model* delivers internally consistent estimates of a wide range of economically interesting targets — prices, merger effects, consumer surplus, productivity.

Question. How reliable are these estimates when the researcher's model may be misspecified? Focus: *multivariate, potentially nonlinear models where the causal variable of interest is endogenous*.

Answer (theory).

- Reliable estimates *require* that the model describes the effects of the endogenous variable approximately correctly.
- Reliable estimates *do not require* the role of the exogenous controls to be correctly specified.
- If the controls are misspecified, reliable estimates require *strong exclusion*: sufficiently many instruments that are unrelated to the controls.

Practice. Engineer strong exclusion in GMM. Illustration: beer demand (Miller–Weinberg 2017); extension to dynamic Cobb–Douglas (App. D.3).

Model and data-generating process

Exclusion (§II.A). Nesting model: $Y_i = \bar{Y}_i(D_i, X_i)$, $D_i = \bar{D}_i(X_i, Z_i)$. Z_i are *excluded variables*, assumed not to causally affect the outcome (cost shifters in demand; input-price shocks in production); X_i are *included variables*, may causally affect the outcome.

Exogeneity (§II.A). Excluded variables unrelated to the unobserved determinants of Y, D . Two flavors:

- Unconditional: $(\bar{Y}_i(\cdot), \bar{D}_i(\cdot)) \perp (X_i, Z_i)$.
- Conditional: $(\bar{Y}_i(\cdot), \bar{D}_i(\cdot)) \perp Z_i \mid X_i$.

Uncond. $\Rightarrow$ cond. Both make Z_i exogenous given X_i ; only uncond. makes X_i itself exogenous.

Researcher's model. Special case of the nesting model: causal effects of D, X on Y governed by a finite $\theta = (\alpha, \beta) \in \mathbb{R}^P$; unobservables captured by mean-zero $\xi_i \in \mathbb{R}^J$, with $Y_i = Y^*(D_i, X_i, \xi_i; \theta)$. $Y^*(\cdot)$ not indexed by i — all unit heterogeneity absorbed by ξ_i .

Invertibility. There is a residual function $R^*(\cdot; \theta)$, determined by Y^* , with $\xi_i = R^*(Y_i, D_i, X_i; \theta_0)$. This is what enables GMM estimation.

Proposition 2. Causally correct $\iff$ for some α_0 , $Y_i(d, x) = Y^{**}(d, x, \xi_i + L_i(x); \alpha_0)$ for some unit-specific $L_i(\cdot)$ and residual ξ_i . Equivalently $R^*(Y, D, X; \theta) = R^{**}(Y, D, X; \alpha) - L^{**}(X; \beta)$: β enters only as an additive X -shift; $\tau(G)$ depends on α alone.

Approximately correct specification, strong exclusion

Distance from causally correct (Def. 3). $\delta(G) = \inf_{\theta} \delta(\theta, G)$ with

$$\delta(\theta, G) = \sum_{j,j'} \mathbb{E}_G \left[\sup_d \left| \frac{\partial \bar{Y}_{ij}(d, X_i)}{\partial d_{j'}} - \frac{\partial Y_{ij}^*(d, X_i, \xi_i(\theta); \theta)}{\partial d_{j'}} \right| \right], \quad \xi_i(\theta) = R^*(Y_i, D_i, X_i; \theta).$$

Causally correct $\iff \delta(G) = 0$. Small $\delta(G)$: some θ matches the causal effects of D on Y ; *controls may enter arbitrarily wrong*.

Strong exclusion (Def. 4). GMM moment $f^* = (f^E, f^I)$ with $\mathbb{E}_G[f^E | X] = 0$ and $\text{rank } \mathbb{E}_G[f^E f^{E'}] \geq \dim \alpha$. f^E identifies α ; f^I identifies β . Overid: see B9 (Lemma 6).

Contrast. Blundell–Bond IVs $f_j^* = (1, K_{i,j}, D_{i,j-1}, K_{i,j-1})'$ depend only on $X \Rightarrow$ fail dynamic strong exclusion (B10). Same failure mode as BLP-style $\bar{X}_{i,-j}$ in the static case.

Main result and the intuition for it

Approximate causal consistency (Def. 5). For any $b > 0$, $\exists \delta > 0$ such that

$$\sup_{\{G \in \mathcal{G}: \delta(G) \leq \delta\}} \sup_{\tau \in \mathcal{T}} |\tau^*(\theta^*(G)) - \tau(G)| \leq b.$$

Proposition (Prop. 3)

If conditional exogeneity holds, any estimator satisfying strong exclusion and strong identification (Def. 6, B9) is approximately causally consistent. Moreover, even if unconditional exogeneity holds, any estimator that is approximately causally consistent must satisfy strong exclusion.

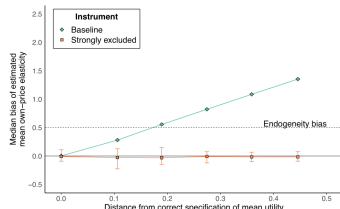
Intuition (linear-residual case).

1. Prop. 2: $Y = Y^{**}(D, X, \xi + L^{\text{true}}(X); \alpha_0)$. Misspecified controls $\Leftrightarrow$ wrong additive shift $g(X) = L^{\text{true}}(X) - L^{**}(X; \beta)$ — a function of X alone.
2. α -moment at α_0 splits: $\mathbb{E}[f^E \xi] + \mathbb{E}[f^E \cdot g(X)]$.
3. Exogeneity kills the first; $\mathbb{E}[f^E | X] = 0$ kills the second (orthogonality to every $g(X)$). Hence $\hat{\alpha} \rightarrow \alpha_0$ regardless of β , and $\tau(G)$ uses α alone.

Recipe and beer-demand evidence (Miller–Weinberg 2017)

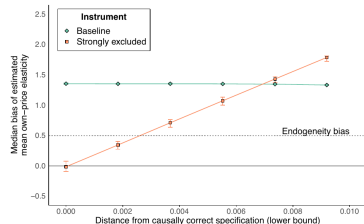
Recipe (§IV.D). Set aside $\dim \beta$ rows of $\hat{f}(X, Z)$ for β ; residualize the remaining $\dim \alpha$ rows: $f^E = \hat{f} - \mathbb{E}[\hat{f} \mid X]$.

(A) Varying the misspecification of mean utility, under causally correct specification



Panel A. CC spec held; x : mean-util misspec.

model of mean utility



Panel B. Mean util misspec; x : $\delta(G)$ lower bound.

y : median bias of $\hat{\alpha}$ (own-price elasticity). MW data (fn. 18): -11.57 strong excl. vs. -4.35 baseline.

B1 • Proposition 1: necessity of approximate causal correctness

Setup. *Causally correct specification* (Def. 3): some θ makes the researcher's model exactly reproduce the causal effects of D_i on Y_i . Distance $\delta(G)$ measures how far G departs from this.

Proposition 1.

- *Negative.* If $\delta(G)$ is unbounded over $\mathcal{G}$, then for every $b > 0$ no oracle $\tilde{\theta}(\cdot)$ achieves $|\tau^*(\tilde{\theta}(G)) - \tau(G)| \leq b$ for all $\tau \in \mathcal{T}$ and $G \in \mathcal{G}$.
- *Positive.* There exists an oracle $\tilde{\theta}(\cdot)$ such that for any $b > 0$, some $\bar{\delta} \geq 0$ gives $|\tau^*(\tilde{\theta}(G)) - \tau(G)| \leq b$ whenever $\delta(G) \leq \bar{\delta}$.

Reading. ACC is necessary for any estimator (oracle or feasible) to deliver bounded error on causal summaries — the floor under Prop. 3's iff. Conclusion holds irrespective of the form of exogeneity assumed on X_i, Z_i .

B2 • Three running examples

Linear (§II.B). $Y_i \in \mathbb{R}$, $Y^*(D_i, X_i, \xi_i; \theta) = \alpha D_i + X_i \beta + \xi_i$; $R^* = Y_i - \alpha D_i - X_i \beta$. Demand reading: $Y = \log Q$, $D = \log p$, $X = \log y$ (isoelastic). $\tau^*(\theta) = \alpha$; $\tau^*(\theta) = \alpha \Delta$ for a price change Δ .

Logit (§II.B). $Y_i \in \mathbb{R}^J$ (market shares),

$$Y_j^*(D_i, X_i, \xi_i; \theta) = \frac{\exp(\alpha D_{i,j} + X_{i,j} \beta + \xi_{i,j})}{1 + \sum_{j'=1}^J \exp(\alpha D_{i,j'} + X_{i,j'} \beta + \xi_{i,j'})}.$$

Residual: $R_j^*(Y_i, D_i, X_i; \theta) = \ln Y_{i,j} - \ln Y_{i,0} - \alpha D_{i,j} - X_{i,j} \beta$, with $Y_{i,0} = 1 - \sum_{j=1}^J Y_{i,j}$ the outside-good share.

Model-implied derivatives:

$$\frac{\partial Y_{i,j}^*}{\partial D_{i,j}} = \alpha Y_{i,j} (1 - Y_{i,j}), \quad \frac{\partial Y_{i,j}^*}{\partial D_{i,k}} = -\alpha Y_{i,j} Y_{i,k} \quad (k \neq j),$$

so $\tau^*(\theta) = \alpha \cdot \frac{1}{J} \sum_j \mathbb{E}_G[D_{i,j}(1 - Y_{i,j})]$ (model-implied avg. own-price elasticity).

Causally correct logit. $\delta(G) = 0$ allows

$$Y_{i,j}(D_i, X_i) = \frac{\exp(\alpha_0 D_{i,j} + L_{i,j}(X_{i,j}) + \xi_{i,j})}{1 + \sum_{j'=1}^J \exp(\alpha_0 D_{i,j'} + L_{i,j'}(X_{i,j'}) + \xi_{i,j'})},$$

$L_{i,j}(\cdot)$ unknown: researcher correctly models the effect of D_i , but $X_{i,j}$ may enter arbitrarily.

Random coefficients logit (§II.B). Unit mass of consumers c per market, $v_{c,i} \sim F(\cdot; \alpha_2)$ mean zero:

$$Y_j^* = \int \frac{\exp(\alpha_1 D_{i,j} + X_{i,j}(\beta + v_{c,i}) + \xi_{i,j})}{1 + \sum_{j'=1}^J \exp(\alpha_1 D_{i,j'} + X_{i,j'}(\beta + v_{c,i}) + \xi_{i,j'})} dF(v_{c,i}; \alpha_2).$$

Invertible (Berry 1994; BLP 1995; BGH 2013): $\xi_i = R^*(Y_i, D_i, X_i; \theta)$.

B3 • Causal summaries for differentiated-goods demand

Definition 1. $\tau(G) = \sum_{j,j'} \mathbb{E}_G \left[\int \frac{\partial Y_{i,j}(d, X_i)}{\partial d_{j'}} d\omega_{i,j,j'}(d) \right]$, weights bounded $\max_{j,j'} \int |d\omega_{i,j,j'}(d)| \leq \bar{W}$.

Average own-price elasticity. $\tau(G) = \frac{1}{J} \sum_j \mathbb{E}_G \left[\frac{D_{i,j}}{Y_{i,j}} \frac{\partial Y_{i,j}}{\partial D_{i,j}} \right]$. Weights $d\omega_{i,j,j'} = \frac{D_{i,j'}}{JY_{i,j}}$ when $j' = j$, $d = D_i$, else 0 (data-dependent).

Cross-price elasticity gap. $\tau(G) = \mathbb{E}_G \left[\frac{D_{i,2}}{Y_{i,1}} \frac{\partial Y_{i,1}}{\partial D_{i,2}} - \frac{D_{i,3}}{Y_{i,1}} \frac{\partial Y_{i,1}}{\partial D_{i,3}} \right]$. Signed weights; cf. Gandhi–Lu–Shi 2023.

Diversion ratio. $\tau(G) = \mathbb{E}_G \left[\frac{\partial Y_{i,2} / \partial D_{i,1}}{\partial Y_{i,1} / \partial D_{i,1}} \right]$. Weights depend on $Y_i(\cdot)$ (Backus–Conlon–Sinkinson 2021).

Counterfactual share change. Price change Δ_i : $\tau(G) = \mathbb{E}_G \left[\int_0^1 \frac{dY_{i,j}(D_i + t\Delta_i, X_i)}{dt} dt \right]$. Uniform weights on $[D_{i,j}, D_{i,j} + \Delta_{i,j}]$.

Consumer surplus change. Smooth functional of $Y_i(\cdot)$ over the price-change path (Remark 2; Nevo 2000a, Town 2001, MW 2017).

B4 • §IV.D: direct procedure for enforcing strong exclusion

Setup. Practitioner has chosen initial instruments $\hat{f}(X_i, Z_i)$ and weight matrix $\hat{\Omega}$ that *do not necessarily* enforce strong exclusion. Assume $L_j^{**}(X_i; \beta)$ is not fully saturated in X_i (else strong exclusion holds automatically).

Direct procedure.

- Set aside exactly $\dim(\beta)$ rows of $\hat{f}(X_i, Z_i)$ — these are f^l , allowed to depend on X_i , pin β .
- Flexibly residualize the remaining $\dim(\alpha)$ rows wrt X_i to obtain f^E that is mean-independent of X_i — pin α .
- At least $\dim(\alpha)$ rows of $\hat{f}$ must depend on Z_i for the procedure to be well-defined.

Intuition. α identified by moments that don't depend on X_i ; β by moments that do. Particularly clear in just-identified case; extends to overid.

Differentiated-goods convention. If $L_j^{**}(X_i; \beta) = L_j^{**}(X_{i,j})\beta$ enters mean utility linearly, including $L_j^{**}(X_{i,j})$ in $\hat{f}$ is natural — its dimension is exactly $\dim(\beta)$, so exclude it from residualization. Inference: B7. Automated alternative: B8.

B5 • Linear special case: Borusyak–Hull (2023) recentered instruments

Setup (BH *Econometrica* 91(6), 2155–2185). $y_\ell = \beta x_\ell + \varepsilon_\ell$; instrument $z_\ell = f_\ell(g; w)$, where g are exogenous shocks and w are predetermined exposure variables (e.g., shift-share, simulated IVs, market-access).

OVB problem (BH Lemma 1). Even when $g \perp \varepsilon \mid w$,

$$\mathbb{E}\left[\frac{1}{L} \sum_\ell z_\ell \varepsilon_\ell\right] = \mathbb{E}\left[\frac{1}{L} \sum_\ell \mu_\ell \varepsilon_\ell\right], \quad \mu_\ell = \mathbb{E}[f_\ell(g; w) \mid w].$$

The *expected instrument* μ_ℓ confounds OLS/IV through non-random exposure.

Recentering (BH Proposition 1). $\tilde{z}_\ell = z_\ell - \mu_\ell$ identifies β under weak shock exogeneity ($\mathbb{E}[\varepsilon \mid g, w] = \mathbb{E}[\varepsilon \mid w]$).

Connection to ABGRS. BH's $\tilde{z} = z - \mu$ is exactly ABGRS's $f^E = \hat{f} - \widehat{\mathbb{E}}[\hat{f} \mid X]$ in the linear case ($g \leftrightarrow Z, w \leftrightarrow X$). ABGRS extends:

- *Beyond linear:* arbitrary $Y^*(D, X, \xi; \theta)$ and vector α .
- *Beyond design-based:* $\widehat{\mathbb{E}}[\hat{f} \mid X]$ via flexible regression when no shock-assignment model exists.
- *Iff direction:* strong exclusion is *necessary* (not just sufficient) for approx. causal consistency on causal summaries.

B6 • Compromises when X is rich or α is vector-valued

When the §IV.D recipe is impractical. Full mean independence $\mathbb{E}[f^E | X] = 0$ on $\dim \alpha$ rows can destroy identifying power. The paper offers two compromises that weaken the guarantee.

Rich X (§V.E; Online App. C.2). Residualize against a coarsening $\chi_j(X_i)$ instead of full X_i . *Choice-set residualization* (χ_j indexes the available product set $\mathcal{J}_i$) and *product-level residualization* ($\chi_j(X_i) = X_{i,j}$) preserve more instrument variance; cost: modest bias appears under causally correct spec when misspecification spans factors outside χ_j (Figs. III, IV, Panel C).

Vector α (§V.F; nested-loop adapt. of App. C.4). For an RC logit with $\alpha = (\alpha_1, \alpha_2)$: residualized $f^{MW,E}$ in an *inner loop* pins only the price coefficient α_1 ; original f^{MW} in an *outer loop* pins β and the RC dispersion α_2 .

Fallback guarantee. Approximate causal consistency for all $\tau \in \mathcal{T}$ no longer holds. *Proposition 5 (Online App. B)* delivers recovery of a restricted α -sensitive class (cf. §IV.C). Fig. V: the average own-price elasticity stays in this class.

B7 • App. C.3: inference under residualization

Two-step semiparametric GMM (App. C.3). Set $f^l = \hat{f}_{1:\dim \beta}$ (kept un-residualized to identify β). Residualize the rest: $f_j^E = \hat{f}_j - \widehat{\mathbb{E}}[\hat{f}_j | X]$ for $j > \dim \beta$ (to identify α).

Parametric first-step. Assume $\mathbb{E}_G[\hat{f}_j(X, Z) | X] = \gamma_j' b_j(X)$ for known basis $b_j(X) \in \mathbb{R}^{k_j}$ (sieve / polynomial / dummies if X has finite support).

1. $\hat{\Gamma} = \arg \min_{\Gamma} \hat{g}(\Gamma)' \hat{\Omega}_g \hat{g}(\Gamma)$, $\hat{g}(\Gamma) = \frac{1}{n} \sum_i b(X_i)' [\hat{f}_{(\dim \beta + 1):L} - b(X_i)\Gamma]$.
2. Stack $(f^l, \hat{f}_{(\dim \beta + 1):L} - b(X)\hat{\Gamma})$ — second component is f^E — and estimate $\hat{\theta}$ by GMM.

Inference: just-id $\rightarrow$ Murphy–Topel (1985); overid + misspec $\rightarrow$ Hall–Inoue (2003) + consistent var.; bootstrap $\rightarrow$ Lee (2014).

Nonparametric first-step. Sieve minimum distance (Ai–Chen 2007, 2012): $h_j(X) = \mathbb{E}_G[\hat{f}_j | X]$ in infinite-dim $\mathcal{H}$, basis $b_j(X) \in \mathbb{R}^{k_{j,n}}$ with $k_{j,n} \rightarrow \infty$. $\hat{\theta} \xrightarrow{P} \theta^*(G)$, asymp. normal, consistent variance.

Remark 7 — debiased GMM (Chernozhukov 2022). Just-identified case: orthogonal moment $\mathbb{E}_G[\tilde{m}_i(\theta, h) + \phi_i(\theta, h, \xi)]$ with influence function ϕ_i ; cross-fitting + ML for nuisance $h(\cdot), \xi(\cdot)$.

B8 • App. C.4: automated nested-loop recipe

Ingredients. Instruments $\hat{f}(X_i, Z_i) \in \mathbb{R}^{L \times J}$, $L \geq P$; weight matrices $\hat{\Omega}^E, \hat{\Omega}^I \in \mathbb{R}^{L \times L}$.

Recipe (Strong exclusion).

- Residualize $\hat{f}(X_i, Z_i)$ wrt X_i via nonparametric regression $\Rightarrow f^E$; define $f^I = \hat{f}$.
- Sample moments $\hat{m}^E(\theta) = \frac{1}{n} \sum_i f^E R$, $\hat{m}^I(\theta) = \frac{1}{n} \sum_i f^I R$ where $R = R(Y_i, D_i, X_i; \theta)$.
- $\min_{\beta} \hat{m}^{I'} \Omega^I \hat{m}^I$ s.t. $\hat{\alpha}(\beta) = \arg \min_{\alpha} \hat{m}^{E'} \Omega^E \hat{m}^E$.

Why it satisfies strong exclusion. Estimand solves a form-(3) moment with $f_G^* = (W_G^E; W_G^I)(f^{E'}, f^{I'})'$, W_G^E zero except upper-left $L \times L$ block. If the regression for f^E is consistent + IVs vary, strong exclusion holds without manual moment partition.

Inference (App. C.4.1). Just-id $\rightarrow$ standard GMM (use C.3). Overid + correct spec:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, (S_0 M_0)^{-1} S_0 \Sigma_0 S_0' (M_0 S_0')^{-1}\right),$$

with block-triangular score S_0 . Efficient outer-loop weight = inverse of asymp. variance of $\hat{m}^I(\hat{\alpha}(\beta_0), \beta_0)$.

B9 • Strong identification, overid, dynamic extension

Strong identification (Def. 6). α strongly identified by f_G^E : for any $\epsilon > 0$ there exists $\zeta > 0$ such that

$$|\mathbb{E}_G[f_G^E R^*(Y_i, D_i, X_i; \alpha, \beta)] - \mathbb{E}_G[f_G^E R^{**}(Y_i, D_i, X_i; \alpha)]| \leq \zeta \implies |\alpha - \alpha^*(G)| \leq \epsilon.$$

Weak-ID handled separately (Andrews REStat 2018).

Overid (Online App. C.1, Lemma 6). Strong exclusion requires $\text{rank}(W_G \Xi_G W_G') \leq \dim \beta$, with $\Xi_G = \mathbb{E}[\mathbb{E}[f^* | X] \mathbb{E}[f^{*'} | X]]$: at most $\dim \beta$ rows of f^* may depend on X .

Dynamic strong exclusion (Online App. D.3; Def. 12, Prop. 8). Panel (i, j) , $X_{i,j} \supseteq (K_{i,j}, K_{i,j-1}, \dots)$. Decompose $W_G f^* = (W_G^E f^*; W_G^I f^*)'$ with $\mathbb{E}_G[W_G^E f_j^* | X_{i,j}] = 0$ and $\text{rank} \geq \dim \alpha \Rightarrow \text{ACC}$ for $\tau_V(G)$.

Counterexample. Blundell–Bond instruments $f_j^* = (1, K_{i,j}, D_{i,j-1}, K_{i,j-1})'$ are fully determined by $X_i \Rightarrow$ fail dyn. strong exclusion (same for OP/LP/Wooldridge/FE). *Fix:* residualize against contemporaneous $X_{i,j}$; productivity innovations, FOCs, and demand-side variation work if mean-independent of $X_{i,j}$.